

Technical Specification of Single Phase Whole Current Smart Meter and Meter Box

This specification covers the following for single phase 5-30A static whole current watt hour smart meters of accuracy class 1.0 with modular pluggable communication modules and integrated load control switches along with pilfer proof meter box to house the meter with its accessories.

1.0 Scope:

- 1.1 Design, manufacture, testing at manufacturer's works before dispatch, packing, delivery and submission of all documents.
- 1.2 The meter shall have bidirectional communication capability through all of the communication technologies defined in IS 16444 and shall communicate with DCU/Gateway Access Point / HES as per the choice /requirement of the utility.
- 1.3 Smart meter shall be with provision of TOD (Time of Day) tariff, also capable of recording and displaying energy in kWh and demand in kW for single phase two wire A.C. loads
- 1.4 Meter shall support on demand data fetching read, remote firmware upgrading, different types of event detection etc.
- 1.5 The meter shall have different import and export registers so that it can be used as net meter also with renewable generation resources.
- 1.6 The meter shall have capability to function in both post-paid and pre-paid mode and any mode can be enabled remotely.
- 1.7 All accessories / hardware required for installation, commissioning and operation for the meter shall have to be provided.
- 1.8 It is mandatory that in case of all manufacturers, the offered meter shall be ISI marked and bidder shall have to furnish valid BIS certification along with the offer.

2.0 Applicable Codes & Standards:

The meters covered in this specification shall conform to the latest editions and amendments of the following Indian/CEA/IEC Standards and shall conform to the regulations of local statutory authorities, unless specified separately elsewhere in this specification.

Sl. No.	Standard	Description
1.	Indian Electricity Act	IE Act 2003
2.	CEA Metering Regulations: 2006	Installation and operation of meters with latest amendments.
3.	CBIP Manual (Pub no.-325)	Standardization of AC Static Electrical Energy Meters
4.	RERC Regulation	On installation and operation of meters dated 29.05.2007.
5.	IS 16444 (Part 1) with latest amendments	AC Static Watt hour Smart Meters, Direct Connected Class 1 and 2
6.	IS- 13779: 1999	AC Static Watt-hour Meters, Class 1 and 2 - Specification
7.	IS- 15884: 2010	Alternating current direct connected static prepayment meters for active energy (Class 1 and 2) - Specification
8.	IS- 11448:	Application guide for AC Electricity meters
9.	IS 15959 Part 1, Part 2 with latest amendments	Data Exchange for Electricity Meter Reading, Tariff and Load Control-Companion Standards
10.	IS15707:2006	Testing Evaluation Installation and Maintenance of AC Electricity Meters.
11.	IS : 9000	Basic Environmental testing Procedures for Electronic & Electrical items.

12.	IS:12346(1988)	Specification for testing equipment for AC Electrical Energy meter
13.	IEC 62052-11	Electricity metering equipments (AC) – General requirements & test conditions Part II metering equipments.
14.	IEC 62053-21	Electricity Metering equipments (AC)- particular Requirements – Part – 21 Static meters for active Energy (class 1 & 2)
15.	IEC – 61000-4-5 (2001-04)	Electromagnetic Compatibility (EMC)-Testing and measurement Techniques, Surge immunity test
16.	IEC 61358:1996	Acceptance inspection for direct connected AC static Watt hour meter for active energy (class 1 & 2)
17.	IEC 62053-61	Electricity Metering Equipment (a.c) Particular requirement- Part- 61 -Power consumption and voltage requirements.

In case of any conflict or discrepancy the order of precedence shall be as follows:

- IS
- IEC
- CBIP technical report-325 (with latest amendments, if any)

In case of any difference between the provisions of these standards and the provisions of this specification, the provisions contained in this specification shall prevail.

NOTE: All requisite tests shall be carried out on the meters which are required to check its conformance to the above-mentioned standards.

3.0 Climatic conditions for installation:

The meters to be supplied against this specification shall be capable of performing and maintaining required accuracy under extreme hot, cold, tropical and dusty climate and solar radiation typically existing in state of West Bengal (India). The meter shall be required to operate satisfactorily and continuously under the following tropical climatic conditions.

Sl No.	Parameter	Value
1.	Maximum ambient air temperature	55°C
2.	Maximum ambient air temperature in shade	45°C
3.	Maximum temperature attainable by the meter	60°C (when exposed to sun)
4.	Minimum ambient temperature	-10°C
5.	Average daily ambient air temperature	40°C
6.	Maximum relative humidity	95%
7.	Number of months of tropical monsoon condition	5 months
8.	Maximum altitude above mean sea level	3000 meters
9.	Average annual rainfall	100 mm to 1500 mm
10.	Maximum wind pressure	200 kg/sq. m
11.	Permitted noise level	45 db

Note: The weather of West Bengal is semi-arid type. Temperatures remain comparatively on the higher end all around the year. Heavy lightning also occurs during the month of June to September.

4.0 Supply/Installation System Data:

Sl No.	Parameter	Value
1.	Type of Installation	Outdoor
2.	System	AC, 1 Phase 2 Wire
3.	Rated Frequency	50 Hz $\pm 5\%$
4.	System Neutral	Solidly Earthed

5.0 General Technical Requirement:

Sl No.	Parameter	Requirement
1.	Meter Type	Single phase, two wire static whole current Watt Hour Smart Meter comprising of measuring elements, display, memory, load switch with pluggable bidirectional communication module all to be housed in pilfer proof meter box.
2.	Connection	Direct (Whole Current)
3.	Reference and Operating Voltage	Reference Voltage (V_{ref}) : 240 V (Ph-N) Operating Voltage: Meter shall be operational with required accuracy from 60% V_{ref} to 120% V_{ref} However, meter shall be capable to withstand with the maximum system Voltage of 433 V (Ph-Ph) for a period of atleast 1 hour
4.	Rated Current	Base Current (I_b) : 5 A Maximum Current (I_{max}) : 30 A
5.	Starting current & Running at no load	Starting Current: The meter should start registering energy at 0.2 % of basic current at unity power factor and first pulse must appear within 10 minutes (i.e. time between two consecutive pulses). Running at no load: When 60% V_{ref} and 120% V_{ref} voltage are applied and no current flows in the current circuit, the test output of the meter must not produce more than one pulse.
6.	Operating Frequency	50Hz $\pm 5\%$
7.	Reference Conditions for testing the performance of the meter	V_{ref} : 240V $\pm 1\%$ Frequency: 50hz $\pm 0.3\%$ Temperature: 27°C $\pm 2^\circ\text{C}$
8.	Accuracy Class	1.0 (shall comply with accuracy requirements under IS 13779)
9.	Meter constant	Imp/ unit (Bidder to specify)
10.	Power Factor Range	Zero lag to Unity to Zero lead
11.	Meter category	Meter shall comply with D1 category of IS 15959 (Part 2)
12.	Power Consumption	Power consumption of the smart meter with integrated communication modules and load control switch shall be as per IS 16444 (Part 1)
	in voltage circuit	Meter shall comply with clause 6.10.1.1 of IS:16444 (Part 1)
	in current circuit	Meter shall comply with clause 6.10.1.2 of IS:16444 (Part 1)
13.	Mechanical requirements	As per clause 6.5 of IS 16444(Part 1)
14.	Calibration	Meter shall be calibrated through software at factory and modification in calibration shall not be possible at site by any means or external

		influence. However, configuration of parameters allowed for field reconfiguration shall be possible through wired or Over the Air (OTA) communication.
15.	Insulation and Impulse test	Meter shall comply with IS 16444 (Part 1)
16.	Minimum Insulation resistance	Minimum Insulation resistance at test voltage $500 \pm 50V$ dc a) Between frame & current, voltage circuits as well as auxiliary circuits connected together: $5 M\Omega$ b) Between each current (or voltage) circuit & each and every other circuit.: $50 M\Omega$
17.	Influence of supply voltage	Meter shall comply with clause 4.4.2 of IS 15884
18.	Short time over current	Meter shall comply with clause 4.4.3 of IS 15884
19.	Immunity to phase and earth fault	Meter shall comply with clause 9.6 of IS 13779
20.	Influence of Self Heating	Meter shall comply with clause 4.4.4 of IS 15884
21.	Influence of heating	Meter shall comply with clause 4.4.5 of IS 15884
22.	Environmental Condition	Meter shall be suitable for environmental conditions as mentioned below
23.	Temperature Range	Operation range: $-10^{\circ}C$ to $60^{\circ}C$
24.	Relative Humidity	0 to 96 %
25.	Resistance against heat and fire	The terminal block and Meter case shall have safety against the spread of fire. They shall not be ignited by thermal overload of live parts in contact with them as per IS 13779.
26.	Resistance against Climatic influence	Meter shall comply with clause 12.6 of IS 13779.
27.	kVAh Calculation	Meter shall be programmed for "Lag only configuration" i.e. lead to be treated as unity PF for kVAh calculation.
28.	Initial start-up of meter	Meter shall be fully functional within 5 seconds after reference voltage is applied to its terminals.
29.	Display	Backlit LCD, minimum 6+1 digits
30.	Communication Capability	Meter shall have the ability to communicate with Head End System (HES) on with any one of the communication technologies mentioned in IS 16444 (Part 1) , i.e, (RF/PLC/ Cellular) in a secure manner. The selection of communication technology shall be as per the site conditions and as per design requirement to meet the performance as per agreed Service Level Agreements (SLAs). In case of Plug-in type communication module, the meter shall log communication module removal /non responsive event with snapshot.
31.	Software	The bidder shall supply software for wired (CMRI/HHU/mobile app and laptop) and remote (preferably web based; for AMI) connectivity. The bidder shall also provide required training of usage of software free of cost.
32.	Communication Layer Protocol	Shall have to comply with clause 9.3 of IS 16444 (Part 1)
33.	Alternative data	In case of meter damage, data stored in meter memory (NVM) shall be

		requirement of technical specification in normal and tamper conditions. n) Top cover and base shall be fixed in such a manner that it shall be break to open type, i.e, ultrasonically or chemically welded. In case any attempt is made to separate the meter cover from the base by using any tools / implements / device, there shall have to be visible evidence of tampering or attempt to open. o) Meter shall have a permanent indication in its display as well as logging of tamper in case of removal of top cover, even in power off condition and it shall not disappear even if cover is re-fitted. It shall have to be treated as a non roll over event.
4.	Fixing Arrangement	a. The manner of fixing the conductors to the terminals shall ensure adequate and durable contact such that there shall be no risk of loosening or undue heating. b. Meter shall have 2 (two) screws in each terminal for effective clamping of cables. c. The screws shall not have pointed ends at the end of the thread. Screw connections transmitting contact force and screw fixing which may be loosened and tightened several times during the life of the meter should be such that the risk of corrosion resulting from contact with any other metal part is minimized. d. Electrical connections shall be so designed that contact pressure is not transmitted through insulating material. e. All terminals and connecting screws and washers should preferably be of tinned / nickel plated brass material. f. The terminals and all connecting screws shall be capable of withstanding a current of 150% of I_{max} atleast for two hours, continuously. g. The meter shall be capable of providing phase to neutral protection up to 433 V for 1(one) hour.
5.	Sealing Arrangement	a) The sealing screws used for the meter cover shall be fixed upside down so that these are tightened from the rear and for screw less design also for fixing the base and cover provision for sealing must be there. b) A run through screw (stud) has to be provided from bottom side & sealing is to be done on the top side of the meter. c) Two independent sealing screws are to be provided at each sides of the meter casing. d) The sealing screws shall be Tinned Brass or Nickel Plated Steel/Brass. In addition to the sealing screws provided to the meter cover, the sealing screws of the terminal cover shall also be Tinned Brass or Nickel plated steel. e) Meters must be supplied with two no. of manufactures' seals fixed between meter base and cover at both sides. If lock /click fit integrated seals are used in sample meters, acceptability of said seal will be decided after through checking.
6.	Ingress Protection	IP 51 or better Meter shall comply requirement of clause 6.9 and 12.5 of IS 13779
7.	Output device	a) Meter shall have flashing LED visible from the front to represent energy (active and reactive) recording. Resolution shall be such that satisfactory accuracy test can be conducted at the lowest load in less than 5 minutes and starting current test in less than 10 minutes. b) Meter shall have provision on LCD for indicating communication status. c) Meter shall have indicator on LCD for displaying the status of load switch.

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8.	Real Time Clock (RTC)	a) The meter shall have internal real time crystal clock to set date and time. b) The Real Time Clock (RTC) shall have long life, at least 10 years. c) RTC shall have separate battery backup. d) Meter shall have capability of time synchronization through optical port/ remote communication with proper authentication.
9.	Battery	Meter shall have Lithium / Lithium ion battery with guaranteed shelf life of 10 years and capacity life of 15 years. In case of battery removal or total discharge same shall not affect the working & memory of the meter even in case of single wire power condition.
10.	Memory	Non-volatile memory (NVM) independent of battery backup, shall retain data up to 10 year without any auxiliary power.
11.	Meter reading in Power Off condition	It shall be possible to read the meter during power off condition. It shall also be possible to read the meter through NIC or with CMRI / Laptop in this condition. If battery is used for the same, it shall have to be a separate battery and not the one used for RTC, i.e., the RTC battery and the battery used for display during power off condition shall not be the same.
12.	Self-Diagnostic feature	The meter shall be capable of performing complete self diagnostic check to monitor integrity of data in memory location all the time. The meter shall have indication for unsatisfactory / non-functioning / malfunctioning of the following: a) RTC b) All display segments c) Battery d) Non-volatile Memory (NVM) e) Synchronization of time of meter & HES clock from MDAS instantly for less than 3 minutes and through HES for more than 3 minutes with alarm at HES.
13.	Load Control Switch	Smart meter shall be equipped with integrated load control switches to control flow of electricity to the load with connect/disconnect commands as per functional need of the system. Load switch shall be in compliance to IS 15884 and IS 16444. Meter shall be remotely settable to support double pole relay for connection/disconnection. The phase and neutral relay shall connect/disconnect on the following conditions: a) Over current b) Load control limit c) Pre-programmed Tamper conditions (like, Meter Cover open detection, Neutral disturbance, Magnetic Interference and single wire etc.). d) Disconnect signal from Utility Control Centre such as balance unavailable in case pre-paid facility is availed by consumer Load Control limits shall be remotely programmable. The reconnection mechanism shall also be remotely programmable. The brief technical particulars of this load switch are furnished below:

Sl No.	DESCRIPTION	Requirement

	retrieval	retrieved using a jig. Bidder need to provide such jigs free of cost (1 jig on each 10,000 meters). Jig shall be designed such that NVM can be plugged easily on it for data retrieval.
34.	Testing as part of Technical Evaluation	Sample meter with all accessories including communication module & load switch and software shall be tested at WBSEDCL's Testing Laboratory as part of technical evaluation of the offered smart meter. Requisite infrastructure needed for simulating the environment beyond the meter testing facilities, i.e, for load switching capabilities, data exchange and communication capabilities are to be provided by the bidder during this testing.
35.	Test and Test Conditions	Test and Test Conditions shall be guided by clause 10 of IS 16444(Part 1). Note: i. The smart meters shall be subjected to specified test for meteorology, for load switching capability, for data exchange and for smart meter communicability. ii. The tests shall include Type Tests, Routine Tests and Acceptance Tests.
36.	Data Retention	AS per CEA Regulations.

6.0 Constructional Requirement:

Sl No	Parameter	Requirement
1.	General	a) The meter shall be compact in design. The entire construction shall be capable of withstanding stresses likely to occur in actual service and rough handling during transportation. The meter shall be convenient to transport and immune to shock and vibration during transportation and handling. b) All insulating material used in the construction of meters shall be non-hygroscopic, non-ageing and of tested quality. All parts that are likely to develop corrosion shall be effectively protected against corrosion during operating life by providing suitable protective coating. c) Meters shall be designed and constructed in such a way so as to avoid causing any danger during use and under normal conditions. However, the following shall have to be ensured: - Personal safety against electric shock - Personal safety against effects of excessive temperature - Protection against spread of fire - Protection against penetration of solid objects, dust & water
2.	Meter Body	a) The meter shall be housed in a safe, high grade, unbreakable, fire resistant, UV stabilized insulating material of good dielectric and mechanical strength, The casing shall be of projection mounting type made up of virgin polycarbonate material. b) The meter cover shall be transparent with viewing window for easy reading of displayed parameters and observation of operation indicators. c) The meter base may not be transparent, but it shall not be black in colour.

3. Other
Constructional
Features

- d) The meter casing shall not change in shape, colour, size, and dimensions when subjected to UV test as per ASTM D 53 for 200 hrs.
- e) It shall withstand 650 °C glow wire test and heat deflection test as per ISO 75.
- f) For testing of changing colour 72 hrs on UV test is applicable.
- g) Meter body shall be sealed in such a way that opening of meter base and cover is not possible without breaking the seals.
- h) Thickness of meter body (Base and Terminal cover) shall be 2mm minimum.

- a) The meter shall be supplied with a transparent extended terminal block cover (ETBC).
- b) The terminal block shall be made of high grade non-hygroscopic, fire retardant, fire resistant material having good dielectric and mechanical strength with brass inserts for connecting terminals.
- c) The bidder shall submit relevant documents regarding the procurement of polycarbonate material. The polycarbonate material of only the following manufacturers shall be used:

- i. G.E. Plastics/SABIC LEXAN 943A, or equivalent like 123R for Top cover & Terminal cover/ LEXAN 503R or equivalent like 143R for base & Terminal Block.
- ii. BAYER Grade corresponding to above
- iii. DOW Chemicals- - DO -
- iv. MITSUBISHI- - DO -
- v. TEJIN- - DO -
- vi. DUPONT- - DO -

- d) The ETBC shall not be easily detachable from the base and be secured to the base using a hinge or with any other suitable arrangement without hinge.
- e) ETBC shall be closed at the bottom to prevent access for wires to terminal holes.
- f) The terminal cover shall have provision for cable entry from the bottom.
- g) The terminal cover shall have sufficient space for incoming and outgoing cable such that these can pass without stressing and damaging the terminal cover.
- h) Diagram of external connections shall be embossed clearly on inside portion of terminal cover.
- i) Meter terminals shall be marked and this shall appear in the diagram.
- j) Clearances shall be as per IS 16444 (Part 1).
- k) All electrically live screws shall be of tinned or nickel plated brass material.
- l) All the fixing holes shall be such designed that once the meter is mounted, the screw heads shall not be accessible.
- m) The meter should be fitted with SHUNT for measuring current in the phase element. The Neutral element may have either C.T. or SHUNT or HALL EFFECT SENSOR with proper isolation. The shunts, used in current circuit must be of high quality having high thermal stability and temperature co-efficient. The shunts shall be E-Beam / Spot welded. In case of Hall Effect Sensor, meter shall record as per

		1	Operating Voltage range	Vref (-40% to +20%).
		2	Operating Current range	IS 16444
		3	Maximum switching power	22 kVA per phase/ per IS 15884 Annex G
		4	No. of poles	Double pole in a single relay
		5	Operation of switches	Simultaneous
		6	Utilization Categories	UC3
		7	Min. number of operation	3000 (close, open each)
14.	Performance requirement for load switching	Utilization category of the load switch shall comply with IS 15884.		
15.	Communication module of meter for AMI	a) Smart meter shall have provision for plugging-in communication module (NIC card) as per clause 1.2(b) of IS 16444(Part 1). b) Meter shall have provision for supporting both the variants of communication module. c) The module shall be able to connect to NAN or WAN (as per the plugged RF or Cellular module) for two-way communication and any of these two will be used at a time. d) Meter shall log communication module removal as an event.		
16.	Meter Sealing Arrangement	a) Reliable sealing arrangement shall be provided to make the meter tamper evident and to avoid fiddling or tampering by unauthorized persons. b) Sealing should be in accordance with IS and CEA metering regulations with latest amendments. c) Approval shall be taken from purchaser for location of seals. d) Adequate sealing provision shall be provided in case of exposed optical port. e) In case of plug in communication module, sealing arrangement shall be provided for the same also.		
17.	Manufacturers / WBSEDCL Seals	a) All meters shall be sealed by the manufacturer at its works with 2 (two) no. of polycarbonate seals with manufacturer's logo and sequential numbers.		
18.	Name Plate and marking	Meter shall have clearly visible, indelible and distinctly marked name plate in accordance with clause 11 of IS 16444 (Part 1).		
19.	Connection Diagram	The connection diagram of the meter shall be clearly shown on terminal cover.		

7.0 Functional Requirement:

S.No.	Parameter	Requirement
1.	Meter category	D1 as per IS 15959 (Part 2).
2.	Data Security	Advanced Security outlined in clause 7.1.2 of IS 15959 (Part 1)
3.	Encryption for data communication	As per clause 7.1 of IS 15959 (Part 2)
4.	Encryption and Authentication For Data Transport	As per clause 7.2 of IS 15959 (Part 2)

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5.	Key Requirements and Handling	As per clause 7.3 of IS 15959 (Part 2)
6.	IP Communication Profile Support	As per clause 8 of IS 15959 (Part 2). Note: Meter shall support TCP-UDP/ IP communication profile for smart meter to HES.
7.	Self- Registration	Last mile mesh network must support auto-registration and self-healing. Meter once powered up should be self-detected and its basic name plate details should be transferred to HES.
8.	List of Parameters for Category D1	As per Clause 11.1 of IS 15959 (Part 2)
9.	Instantaneous Data	As listed in Table A1 of IS 15959 (Part 2)
10.	Billing Profile Data	Billing profile parameters are as listed in Table A4 of IS 15959 (Part 2) Notes: a) At the end of each billing cycle, meter shall generate and store in memory parameters as per provisions provided in clause no. 14 of IS 15959 (Part 2). b) Support for selective access shall be provided for billing parameters as per clause no 11.3 of IS 15959 (part 1). c) The current cycle billing parameters shall be readable as the values of the latest billing period, on demand. This shall be in addition to the last 6 billing period data which shall be available in the profile buffer as the last 6 entries in the buffer.
11.	MD reset	24:00 Hrs. of last day of each month.
12.	Billing period counter	Cumulative billing period counter since installation and available billing periods shall be provided as per clause 11.2 of IS 15959 (Part 1).
13.	Selective access of billing data	By entry.
14.	Billing period reset mechanism	As per clause 10 of IS 15959 (Part 1)
15.	MD Registration	Block window with programmable integration period as per requirement. (Default period : 30 min)
16.	Load Data survey/ Interval Data	As per IS15959 (Part 2) Load Survey data shall be measured and recorded at the end of each interval for last 60 days.
17.	Daily Load Profile	As detailed in Table A3 of IS 15959 (Part 2) The parameters listed in this table shall be logged at midnight (00:00 hrs) everyday and data shall be preserved for last 60 days.
18.	Name Plate Details	As detailed in Table A12 of IS 15959 (Part 2)
19.	Programmable parameters	As detailed in Table A13 of IS 15959 (Part 2) Note: The parameters can be programmed remotely by HES and locally by authorized devices (CMRI, laptop) through serial

		communication port via proper access rights. Every transaction shall be logged in non-volatile memory of the meter with date and time stamp.
20.	TOD Metering	As per clause 9 of IS 15959 (Part 1) The meter shall have facilities to record active energy, apparent energy and MD in at least 8 time zones. The time zones shall be user programmable remotely through HES and locally by authenticated MRI/Laptop/RMR command. Necessary software for the same is to be provided by the bidder. At present TOD timings will be programmable as follows: TOD 1: 06:00 Hrs. to 17:00 Hrs. TOD 2: 17:00 Hrs. to 23:00 Hrs. TOD 3: 23:00 Hrs. to 06:00 Hrs.
21.	Push Services	As per clause no. 6 of IS 15959 (Part 2) Note: There shall be a provision for the smart meter to automatically notify data, events, and messages to a destination client system in an unsolicited manner (without a request from a client).
22.	Periodic push (SM to HES)	As per clause no. 6.1.1 of IS 15959 (Part 2) Note: - In this service the selected objects list is to be notified periodically by the smart meter to HES. - Meter shall be able to push the following objects including some instantaneous parameters to HES at predefined intervals: - Device ID - Push setup ID - Real Time clock- Date and time - Voltage - Phase current - Signed Power factor - Apparent power KVA - Active power Kw - Cumulative Active Energy kWh (Import) - Cumulative Apparent Energy kVAh (Import) - Cumulative Active Energy kWh (Export) - Cumulative Apparent Energy kVAh (Export) - Other attributes as per IS 15959 (Part2) i.e. Send Destination, Communication Window, Randomization Time Interval, number of retries and repeat delay shall be decided in the event of manufacturing.
23.	Event push (SM to HES)	As per clause no. 6.1.5 of IS 15959 (Part 2) Note: In this service the smart meter is to report to HES status change of any of the identified events.

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24.	Firmware upgrade	As per clause 9 of IS 15959 (Part 2) Note: i. Smart meter shall support remote firmware upgrade feature ii. Firmware upgrade shall be limited to the communication firmware only. iii. Firmware upgrade shall use the Image transfer classes and mechanisms specified in IEC62056-6-2 and IEC62056-5-3.
25.	Disconnection mechanism	As per clause 11.1 of IS 16444 (Part 1) Note: i. List of events for disconnection to be pre-programmed shall be provided by utility. ii. Load Control limits shall be programmable locally and remotely. iii. Meter shall use the disconnection control object as defined in clause 10 of IS 15959 (Part 2). iv. Relay for connect/disconnect shall comply all relevant requirements of IS 15884 **Disconnection Logic must be programmable at the time of installation
26.	Reconnection mechanism	As per Clause 11.2 of IS 16444 (Part 1) Note: i. Reconnection shall be done from HES except for over current and load control limit. In case of failure of communication / HES, reconnection shall be possible through Handheld Device (CMRI/mobile app) locally via proper security. ii. Reconnection in case of prepayment meter shall be as per prepayment profile and balance/credit availability in the meter. **Re-connection Logic must be programmable at the time of installation
27.	Load switch event logging	As per clause 11.5 of IS 16444 (Part 1) Note: Meter shall log with date time and other relevant parameters, all connections and disconnections as events.
28.	Status of Load Switch	As per clause 11.4 of IS 16444 (Part 1) Note: Indication of status of load switch (i.e., connected/disconnected) shall be available on display as well as at HES. Connection and disconnection should be logged as events.
29.	First breath and last gasp	As per clause 11.7 of IS 16444 (Part 1) Smart Meter shall detect "First breath"(power on) and "Last gasp"(power off) condition and communicate to HES.
30.	On demand readings	On request from HES
31.	Schedule for meter readings	Programmable through HES

8.0 Anti-tamper and Fraud Detection Requirement: The meter shall offer a link less design i.e. there shall be no isolation link provided between the current and voltage circuits and hence there would not be any possibility of tampering with the same. Tamper/ Event recoding shall follow IS 15959 (Part-2) and would log the event and send alarm at Head End System. The meter shall be capable of recording the following tamper events in memory.

Event Logging Details:

1.	Current Related	As listed in table A5 of IS 15959 (Part-2)
2.	Earth Tamper	In case of earth tamper, apart from logging the event, continuous indication through LED flag or icon on display shall have to be provided
3.	Power Related	As listed in table A6 of IS 15959 (Part-2/3??)
4.	Power Off/On	Meter shall log Power Off event occurrence if the phase voltage is absent beyond a threshold period.
5.	Low Voltage Event Logging	Meter shall log Low Voltage Event in case the voltage reaches below a threshold value beyond a threshold period.
6.	Protection against HV spark	In case a spark of up to 35 kV is applied externally to the meter including its communication port with its cover in place, using a spark gun / ignition coil, meter shall either remain immune & continue to record energy with allowable accuracy or log the event.
7.	Others	As listed in table A8 of IS 15959 (Part-2)
8.	Neutral disturbance	Meter shall log all events when AC/DC Pulsating Voltage is injected in neutral circuit especially when it can affect the recording of energy.
9.	External Magnetic Interference	a) The meter shall either remain immune to tamper through application of external magnetic field (AC electromagnet or DC magnet) as per value specified in CBIP 325 or if the metering gets affected then meter shall record energy with I_{max}, rated voltage and unity P.F. as per CBIP 325 and same shall also be logged as event with date & time. b) In case of abnormal permanent magnetic field, either meter shall remain immune or if the metering gets affected then meter shall record energy with I_{max}, rated voltage and unity P.F and it shall also be logged as event with date & time. c) The meter shall be provided with built in logic/ indication and sensor to detect event as tamper beyond meter's magnetic immunity level and also display such occurrences. The meter accuracy or shall not be affected by permanent magnetic field up to meter's magnetic immunity level. d) At the time of restoration of magnetic tamper the event shall be logged with actual parameters.
10.	Single wire power	Meter shall be immune (error within $\pm 4\%$) and shall start energy recording for connected load above 0.5 A
11.	Non rollover events	As listed in table A9 of IS 15959 (Part-2) For these events, only date and time shall be captured.
12.	Top cover open	Meter shall detect opening of top cover and this event shall be logged as non roll-over event. This even shall not have any associated restoration.
13.	Connection Related	The meter shall not get affected by any remote control device specified

	Tamper Conditions	under CBIP 325 & shall continue recording energy.
14.	I/C & O/G Interchanged	Meter shall record forward energy within acceptable limits of error specified for accuracy class 1.0. (IS?) Note: Not applicable for Import-Export Mode
15.	Phase & Neutral Interchanged	Meter shall record forward energy within acceptable limits of error specified for accuracy class 1.0. (IS?) Note: Not applicable for Import-Export Mode
16.	I/C (Phase & Neutral) Interchanged, Load Connected to Earth	Meter shall record forward energy within acceptable limits of error specified for accuracy class 1.0. (IS?) Note: Not applicable for Import-Export Mode
17.	I/C Neutral Disconnected, O/G Neutral & Load Connected to Earth	Load switch of the meter shall disconnect the load to avoid load drawing under this condition
18.	I/C Neutral Disconnected, O/G Neutral Connected to Earth Through Resistor & Load Connected to Earth	Load switch of the meter shall disconnect the load to avoid load drawing under this condition
19.	I/C Neutral connected, O/G Neutral Connected to Earth Through Resistor & Load Connected to Earth	Meter shall record forward energy
20.	Event Logging	a. Minimum 200 no. of events with date and time stamp preferably along with snapshots of V, I, PF and kWh shall be retained in memory. b. The logging will be on FIFO basis. c. Occurrence and restoration are considered as separate event d. For each occurrence event captured, the cumulative tamper count shall be incremented. e. Selective access shall be provided as per clause 11.3 of IS 15959 (Part 1).
21.	Parameter Snapshot	Parameters as per table A11 of IS 15959 (Part 2) are to be captured during occurrence and restoration of logged events.
22.	Tamper Indication	Appropriate Indications/Icons for all tampers shall appear on the meter display either continuously or in auto display mode.
23.	Tamper Logics	As per Annexure-II
24.	High Frequency Jammer	Meter shall have to be immune on this test.

12	Display Power Up in Absence of Main Power	The meter shall have the provision of providing the display of billing parameters (Auto Display) in absence of main supply. Press of push button shall activate the display to facilitate hands free meter reading with auto-off provision. Performance of battery backup must be as per requisite criteria specified elsewhere in this technical specification.
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10.0 Software & Communication:

1.	Communication Ports	Communication port required in meter are as follows.
	Optical	Meter shall have one optical port. It shall be compatible for data transfer over RS 232 standard
2.	Plug-in Communication Module (NIC)	The smart meter shall have a dedicated sealable slot for accommodating plug-in type bi-directional communication module which shall integrate the respective communication technology (RF/PLC/ Cellular) with the smart meter and act as interface between the meter and HES.. The plug-In module shall be field swappable/ replaceable.
3.	Integration	The bidder must ensure bi-directional data communication between the meter and HES for both RF mesh or cellular communication technology.
4.	Software support and	a) The bidder shall supply following software and provide required training & manuals to use the same free of cost : I. Software for local communication, i.e, for HHU/CMRI/Mobile App and Laptop/PC. This software can be Android or Windows OS based. II. Software for firmware upgrading from remote and mass deployment. b) Bidder shall ensure integration of software with any of WBSEDCL system during the life of meter free of cost. WBSEDCL will provide all the required support for integration activity. c) The bidder shall have to ensure DLMS compliance of the meter for both local & remote communication through its Optical port and NIC card.
5.	Software for local communication	The manufacturer has to provide software capable of downloading all the data stored in meter memory through CMRI/HHU/Mobile app/Laptop/PC. The manufacturer shall also provide software for android or windows based HHUs along with software for traditional CMRI/HHU devices. The bidder has to ensure that all required reading and diagnostic tools are available with them to ensure SLA as mentioned in Tender. Specification and costing of these tools shall be shared with WBSEDCL in case WBSEDCL wants to maintain their own devices.
6.	Training	Manufacture shall impart training to WBSEDCL personnel for usage of software
7.	Port protection	All ports shall be optically isolated from the power circuit.
8.	Operation	Optical port and NIC Card shall work independently. Failure of one (including display) shall not affect the working of the other.
9.	Communication protocol	As per IS 15959 (Part 2). Other protocols shall not be acceptable.
10.	Data transfer rate	Communication ports shall have to support data transfer rate of 9600 bps(minimum).

9.0 Meter Display:

The measured value(s) shall be displayed on a Liquid Crystal display (LCD) register. The display shall have backlit capability for easy reading from meters. When the LCD is placed at a constant temperature of 65°C for a period of 30 minutes in operating condition and 80°C for 30 minutes under de-energized / storage condition, it shall not get deformed.

1.	LCD Type	STN/HTN Liquid crystal with backlit capability
2.	Viewing angle	a) Display shall have 35 degree up & down viewing angle from eye level. b) The display visibility shall be sufficient to read the meter mounted at a height of 0.5 m as well as at the height of 2 m.
3.	Display of Meter Serial No	Every meter shall have a unique alpha-numeric serial no. for displaying either in 9 complete places or in billboard fashion in "Separate Scrolling Mode". The meter serial number shall also be programmed into meter memory for identification through communication port.
4.	Register	The register shall be able to record and display starting from zero, for a minimum of 2500 hours, the energy corresponding to rated maximum current at reference voltage and unity power factor. The register shall not roll over in between this duration. The kWh energy registration shall take place with at least 6+1 digit.
5.	LCD language	English
6.	Display Mode: Auto	The list of parameters to be displayed in auto mode shall be provided separately for prepaid meter, postpaid meter, export – import meter, meter with TOD facilities etc. The display time for each parameter and the auto cycle duration shall be programmable. The parameters shall be displayed sequentially as per specified list.
7.	Display Mode: Push Button	The list of parameters to be displayed in push button mode shall be provided separately for prepaid meter, postpaid meter, export – import meter, meter with TOD facilities etc. The parameters shall be displayed sequentially as per specified list. Instantaneous, cumulative, periodic data as well as history data, event details. Etc. shall be listed in each type of meter specified above.
9.	Permanent Display on LCD	a) Supply Indication b) NIC Card Status c) Relay Status d) Earth Load Indication e) "Meter Cover Open" Tamper event
10.	Display indications	a) Appropriate indications/flags for all tampers and self-diagnostic features b) Data Communication c) Relay Status
11.	High Resolution Display	The meter shall also be capable of offering a high resolution display which shall enable conducting of dial testing in the shortest possible time and as a minimum, the meter shall be capable of offering a resolution of 4 digits after decimal & 2 digits before decimal for the high resolution kWh or 2 digits after decimal and 4 digits before decimal for the high resolution Wh, display.

11.0 Marking of Meter: The marking on the meter shall be in accordance with relevant clauses of IS 16444 and IS 13779. Colour of the Marking Plate will be Light Grey. The basic marking on the meter nameplate shall be as follows (all other markings as per IS 15959A2_R1 clause E - 10.1 shall also be there):

1. Manufacturer's name & trade mark
2. Place of manufacture
3. Type Designation
4. No. of phases & wires
5. Meter serial number
6. Month & year of manufacture
7. Reference voltage
8. Rated current
9. Operating frequency
10. Principal unit(s) of measurement
11. Meter constant (imp/kwh)
12. Class index of meter
13. "Property of WBSEDCL"
14. Purchase Order No. & Date
15. Bar coded meter serial number along with month & year of manufacture
16. BIS marking
17. Firmware Version
18. Communication technology for WAN or NAN (with carrier frequency)
19. Symbol of load switch

12.0 - Guaranteed Technical Particulars:

The bidder shall furnish all the necessary information as desired in the Schedule of Guaranteed Technical Particulars and data, appended with this Specification. If the bidder desires to furnish any other information in addition to the details as asked for, the same may be furnished against the last item of this Annexure-I.

13.0 Technical Deviations:

Any deviation in Technical Specification as specified in the Specification shall be specifically and clearly indicated in a Deviation Format.

14.0 Testing & Test Facilities At Manufacturer's Place:

14.1 The manufacturer shall have to provide or arrange facilities for conducting all the tests specified in IS 16444 (Part 1) for pluggable type smart meters. The meter shall be subjected to tests for metrology, for load switching capability, for data exchange protocol and for smart meter communicability. The test for metrology shall include the "Type Tests, Routine Tests and Acceptance Tests" identified in IS 13779.

14.2 The bidder shall indicate the details of the equipment/instrument/tools available at the manufacturer's premises for conducting the requisite tests as per relevant Standards. The bidder shall indicate the sources of all equipment/ instruments.

14.3 The standard meters used for conducting tests, along with all other required equipment/instruments shall be calibrated periodically at any NABL Accredited Test Laboratory and test certificates shall be made available at manufacturer's works for verification of purchaser's representative.

15.0 Sample Meter Testing At WBSEDCL's Testing Laboratory:

- 15.1 Sample meter with all related accessories including communication module & load switch and software shall be tested at WBSEDCL's Testing Laboratory as part of technical evaluation of the NIT.
- 15.2 The bidder will have to submit sample meters along with pilfer proof meter box and all the accessories including requisite software in sealed casing / carton along with relevant documents as per provision laid down in NIT.
- 15.3 Requisite infrastructure needed for simulating the environment beyond the meter testing facilities, i.e, for testing of load switch functioning, local & remote bi-directional data exchange and communication capabilities are to be provided by the bidder during this testing.

16.0 Inspection:

- 16.1 The purchaser may carry out the inspection at the works of the meter manufacturer at any stage of manufacture. The manufacturer shall grant free access to the purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any meter manufactured as per this specification shall not relieve the supplier of his obligation of furnishing the meters in accordance with the specification and shall not prevent subsequent rejection if any meter is found to be defective.
- 16.2 All acceptance tests and inspection shall be made at the place of manufacturer unless otherwise especially agreed upon by the bidder and purchaser at the time of purchase.
- 16.3 The Bidder shall provide all reasonable facilities without charge to the inspector, to satisfy him that the meter under reference is being furnished in accordance with this specification.
- 16.4 The supplier shall keep the purchaser informed in advance, about the manufacturing programme for each lot so that arrangement can be made for inspection.
- 16.5 The purchaser reserves the right to insist for witnessing the acceptance / routine testing of the bought out items. The supplier shall give at least 15 days' advance intimation to enable the purchaser to depute his representative for witnessing the acceptance and routine tests.
- 16.6 The purchaser reserves the right for type testing of any meter & meter casing etc. from any of the offered lots, received at any destination stores.

17.0 Quality Assurance Plan:

The design life of the meter shall be minimum 20 years and to prove the design life the firm shall have at least the following quality Assurance Plan:

- The factory shall be completely dust proof.
- The testing rooms shall be temperature and humidity controlled as per relevant standards.
- The testing and calibrating equipments shall be automatic and all test equipment shall have their valid calibration certificates from NABL accredited laboratory.

- d. Power supplies used in testing equipment shall be distortion free with sinusoidal waveforms and maintaining constant voltage, current and frequency as per the relevant standards.

During the manufacturing of the meters the following checks shall be carried out.

- i) Meter frame dimensions tolerances shall be minimum.
- ii) The assembly of parts shall be done with the help of jigs and fixtures so that human errors are eliminated.
- iii) The meters shall be batch tested on automatic, computerized test bench and the results shall be printed directly without any human errors.

The Bidder shall invariably furnish the following information along with his bid, failing which his bid shall be liable for rejection. Information shall be separately given for individual type of material offered.

- Statement giving list of important raw materials
- Names of sub-suppliers for the raw materials
- List of standards according to which the raw materials are tested
- List of tests normally carried out on raw materials
- Information and copies of test certificates in respect of bought out accessories
- List of manufacturing facilities available
- Level of automation achieved
- List of areas where manual processing exists
- List of areas in manufacturing process, where stage inspections are normally carried out of quality control and details of such tests and inspections
- List of testing equipment available with the bidder/manufacturer for final testing of equipment specified
- Test-plant limitations, if any, vis-à-vis type, special acceptance and routine tests specified in the relevant standards and this specification. These limitations shall be very clearly brought out in schedule of deviations.

The manufacturer laboratory must be well equipped for testing of the meters. They must have computerized standard power source and standard equipment calibrated not later than a year (or as per standard practice). The details of testing facilities available for conducting

- a) The routine tests.
- b) Acceptance tests shall be furnished with the bid.

18.0 Guarantee: As per NIT

19.0 Replacement Of Defective Meters: As per NIT

18.0 Guarantee: As per NIT

19.0 Replacement Of Defective Meters: As per NIT

ANNEXURE - I

GUARANTEED TECHNICAL PARTICULARS FOR SINGLE PHASE-2 WIRE SMART ELECTRONIC ENERGY METER OF ACCURACY CLASS 1.0

S. No.	Particulars	Remark Bidder	by
1.	Makers Name & Company		
	a) Meter Serial No.		
	b) Manufacturer Name		
	c) Firmware Version for meter		
	d) Firmware Version for communication module		
	e) Year of manufacturer		
	f) Software for smart meter remote configuration		
	g) Web Based Software for Meter Data		
2.	Model		
3.	Type of Meter		
4.	System voltage with variations		
5.	Standard to which the meter conforms		
6.	Current		
	a) Basic current (i_b)		
	b) Minimum starting current		
	c) Current overloading capacity		
	d) Short time over current		
	e) Maximum current I_{max} continuously with accuracy		
7.	Frequency with variations		
8.	Humidity		
9.	Temperature		
10.	Altitude		
11.	Class of index		
12.	Accuracy		
	a) Current range		
	b) Voltage range		
	c) Frequency range		
	d) Temperature range		
	e) PF range		
13.	Demand & integration Period		
14.	Specific Dynamic Range		
15.	Specified Working Range		
16.	Pulse output		
17.	Register (Electronic)		
18.	Internal Meter Multiplying Factor		
19.	Terminal Connection		
20.	Meter earthing		
21.	Power loss in each current circuit at basic current in VA & watt		
22.	Power loss in each voltage circuit at reference in VA & watt		

S. No.	Particulars	Remark Bidder	by
23.	Display device a) Type of display i.e. LCD b) Character size of display digits c) No. of display digits for data d) No. of display digits for parameter identification e) Life of display unit (guaranteed) f) Method adopted for display overflow g) Indication of healthiness of potential & current h) Provision of Latching Relay connection /disconnection status i) Earth Load Indication (if condition occurred) j) Meter cover open Tamper Event		
24.	Maximum Demand a) Parameters available b) Integration period		
25.	Provision for MD reset a) Communication driven reset b) Auto reset at 24:00 hrs at the end of each billing cycle c) Type of MD computation		
26.	Display parameters in auto scrolling mode LCD Test Credit Balance in INR Relay status Meter Sr. No. Date Time Cumulative kWh Current month MD Instantaneous Voltage Instantaneous Current Instantaneous Load KW Tamper count		
27.	Load Survey Parameters for last 60 days Real Time Clock – Date & Time Current(I) Voltage (V) Block Energy – kWh-(Import) Block Energy –kVAh-(Lag) Block Energy –kVAh-(Lead) Block Energy –kVAh-(Import) Block Energy –kWh-(Export) Block Energy –kVAh-(Export)		
28.	Programmable Parameters a) Real Time Clock - Date and Time b) Demand Integration Period c) Profile Capture Period d) Single – action Schedule for Billing Dates e) Activity Calendar for Time Zones etc., f) Time Zones script table		
29.	MD integration		

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S. No.	Particulars	Remark Bidder	by
	a) Integration period of MD (Minutes)		
	b) Principle of operation		
30.	Overall dimensions, weight & drawing		
31.	Reference standards		
32.	No. of digits displayed		
33.	Parameters read out by MRI / HHU		
	a. Meter serial number, Model, Make		
	b. All parameters as specified in the bid document		
	c. Load Survey data.		
	d. Tamper events details of at least 200 records (in and out) with date and time , (occurrence & restoration are considered separate event)		
	e. Self-diagnostic details (Real time calendar, low battery)		
34.	Communication interface available		
	a) For calibration		
	b) For data transfer		
35.	Non-volatile memory retention time in absence of power		
36.	Provision for connectivity (RS 232)		
37.	Max error due to variation in		
	a) Voltage Variation - 15% to + 10%		
	b) Voltage Variation - 40% to + 20%		
	c) Current 2% to 600% of rated basic current		
	d) Frequency - +/- 5%		
	e) Temperature - + 70c		
	f) PF (0.0 lag -UPF-0.0 lead)		
38.	Anti-Tamper Features		
39.	Power Supply back up		
	a) For sorting recorded values		
	b) For taking reading		
	c) For downloading data		
40.	Measuring principle employed for		
	a) KWH, KVAH		
	b) Maximum demand parameters		
	c) Power Factor		
41.	Guarantee period offered for		
42.	Microprocessor		
	a) Address in bits		
	b) Sampling rate		
43.	Tamper and fraud proof provision for		
	a) Meters		
	b) Software		
44.	Sealing arrangement provided		
	a) Meter body		
	b) Meter Terminal block		
	c) Communication Port		
45.	Degree of protection against dust, moisture etc.,		
46.	Details of battery indication		
	a) Guaranteed life of battery		

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S. No.	Particulars	Remark Bidder	by
	b) Low battery indication		
	c) Internal battery		
47.	Provision of real time clock		
48.	Self-diagnostic features		
49.	Facilities for conducting acceptance test and routine test in factory with additional acceptance & other acceptance tests.		

ANNEXURE-II

Sl. No.	Tamper	Occurrence Threshold	Occurrence Time	Restoration Threshold	Restoration Time	Remarks	Mode
1	Reversal	$V_x > 60\%$ of V_{ref} and $< 115\%$ of V_{ref} $I_x > 10\%$ of I_b (direction is negative) $P.F. > 0.5$	3 minutes	$V_x > 60\%$ of V_{ref} and $< 115\%$ of V_{ref} $I_x > 10\%$ of I_b (direction is positive) $P.F. > 0.5$	3 minutes		Import
2	Single Wire/Neutral Missing	$I_x > 20\%$ of I_b (Maximum) $V_x < 40\%$ of V_{ref}	3 minutes	$I_x > 1\%$ of I_b $V_x > 50\%$ of V_{ref}	Immediate	Recording of energy must be started within 1 minute.	Import & Export
3	Neutral Disturbance	As per bidders' tamper logic.	3 minutes	As per bidders' tamper logic.	3 minutes	Recording of energy must be started within 1 minute.	Import & Export
4	Magnetic Tamper	As per bidders' tamper logic.	1 Minute	As per bidders' tamper logic.	1 Minute		Import & Export

5	Earth Tamper	Iph-In> 6% of Imax(maximum between phase and neutral current) Wattage>24 watt Vx> 60% of Vref And <115% of Vref P.F.>0.5 On Removal of Meter Cover the Meter will lock Cover Open Event with Date and Time Stamp, It must be reflected in Auto Display mode	3 minutes	Iph-In< 6% Vx> 60% of Vref And <115% of Vref P.F.>0.5	3 minutes	Indication is Must. Logging may or may not be provided.	Import & Export
6	Cover Open	On Removal of Meter Cover the Meter will lock Cover Open Event with Date and Time Stamp, It must be reflected in Auto Display mode	Immediate	No restoration shall be allowed without due authorization	Immediate	Indication is Must and Logging should be provided	Import & Export

TECHNICAL SPECIFICATION FOR PILFER PROOF METER BOX SUITABLE FOR SINGLE PHASE STATIC SMART ENERGY METER

1. SCOPE:

This specification covers manufacture and supply of Pilfer Proof Meter Box (PPMB) suitable to house Single Phase Static Smart Energy Meter along detachable communication module and all other accessories.

The Meter Box shall be wall mounted type with ability to protect meter with its assembly against harsh weather. The box shall be anti-corrosive, dust proof, shock, vermin & water proof, pilfer proof, fire proof and UV stabilized. The enclosures shall not deform or melt when exposed to fire.

2. TECHNICAL REQUIREMENT:

The Meter Box i.e base and cover shall be made of hot press/injection moulded, unbreakable, high grade fire retardant Engineering Plastic (Acrylonitrile Butadiene Styrene)/Polycarbonate, with minimum thickness 3.0 mm having good dielectric and mechanical strength. The material must be such that the Meter Box shall not change in colour, shape, size, dimensions when subjected to Ageing Test.

The Meter Box shall have top tapered surface and round corners to prevent any water logging on the top of meter cover. The overall dimensions of the enclosure shall be suitable for housing the single phase smart meter along with all its attachments/accessories as offered by the bidder and there shall be a clearance of 30 ± 2 mm from top & both sides of the meter. Bottom side clearance shall be 75 ± 5 mm from the lower edge of terminal block of the meter. Clearance from front side and back side of the meter shall be 15 ± 2 mm and 10 ± 2 mm respectively.

The Meter Box shall be capable of withstanding the mechanical, electrical and thermal stresses well as the effects of humidity which are likely to be encountered in service. At the same time the box shall ensure desired degree of safety. The material used shall be adequately stabilized against detrimental effect of light and weather. The surface appearance of the moulded parts must be smooth, non-porous and homogeneous, free of ripples, defects and marks. No fillers or fibres shall be visible at any place.

The box shall comply in all respect with the requirement of latest amendments of IS /ASTM. Applicable degree of protection shall be IP 42 or better.

All accessories like nuts, bolts, washers etc. shall be galvanized.

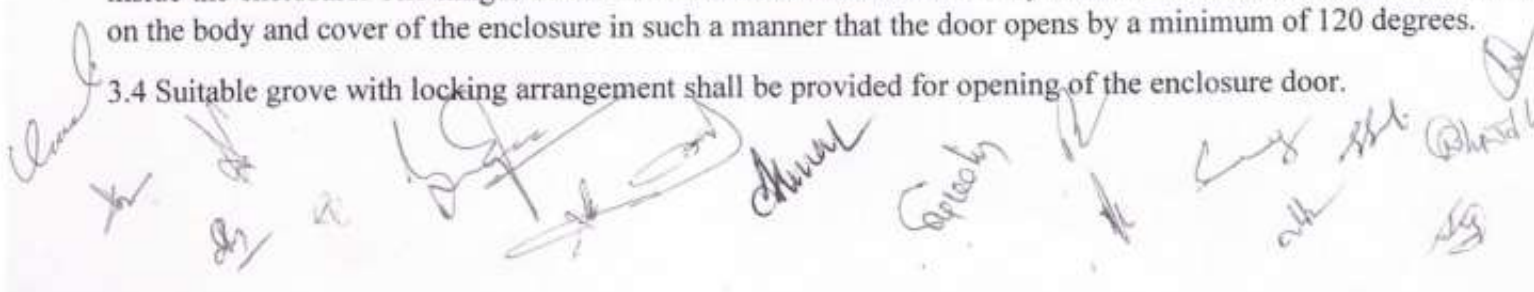
3. CONSTRUCTION:

3.1 The enclosure shall be single piece moulded with hot process compression moulding or injection moulding.

3.2 Dimension: To be specified by the bidder.

3.3 Hinge: A minimum of 2 nos. brass/stainless steel hinges in left side of the door and 1 no. brass/stainless steel hinge/stainless steel U type latch with locking arrangement in right side of the door shall be provided inside the enclosure. The hinges of the door shall be concealed and they shall be fixed to the flanges provided on the body and cover of the enclosure in such a manner that the door opens by a minimum of 120 degrees.

3.4 Suitable groove with locking arrangement shall be provided for opening of the enclosure door.



3.5 Fixing arrangement: The meter box shall have sufficient number of mounting brackets made out of same material as of meter box with provision for hole of suitable diameter for mounting the box on the wall or on wooden board with suitable screws.

3.6 Any other suitable arrangement for fixing the box may be accepted after verification.

3.7 Sealing arrangement: The Box shall have provision for sealing of the base with the cover to make it fully tamper proof.

3.8 Incoming and outgoing cable arrangement: Required number of holes at bottom with suitable diameter shall be provided in the box for cable/wire entry. The holes must be totally covered with neoprene rubber gasket (NRG). NRG will be punched for entry of incoming and outgoing cable/wire.

3.9 Base and Cover details: Thickness of the meter box shall not be less than 2.0 mm on all sides including door. The meter box shall have 4 mm thickness of the tongue and groove area. The meter box cover shall be made overlapping type having collars on all four (4) sides and shall be provided with Neoprene rubber gasket of minimum 2.5 mm dia to completely fit in the grooves of the base. The base of the meter box must have a groove to hold the gasket and the overlap of the top cover with base shall be sufficient. The tongue of the base shall ensure proper sealing arrangement against ingress of rainwater and dust inside the box.

3.10 The Enclosures shall generally comply with the provision of IS 14772 or IEC 695. The enclosures shall be suitable for outdoor application. The enclosure shall be with good workmanship.

3.11 Soft neoprene/nitride rubber gaskets shall be provided all round wherever required for protection against entry of dust and water. The gasket shall conform to Type-III as per IS-11149. The enclosure shall comply with IP-42 or better degree of protection.

3.12 The Enclosures shall be of crystal clear (transparent) polycarbonate.

3.13 Marking/Embossing: The following information shall be clearly & indelibly embossed/laser printed on the cover and base of the Meter Box. No printed paper sticker will be accepted. The top & bottom corner of Meter Box Sl. No. shall be same for the particular Meter Box.

- a) Property of WBSEDCL
- b) Name/Brand name of Manufacturer
- c) Meter Box Sl. No. (Embossed / laser printing on both the base and covers of Meter Box)
- d) Sign of Danger

3.14 Submission of Sample: The bidder shall submit a sample Meter Box as per specification along with the sample meter as per provision laid down in NIT.

3.15 Type testing including material identification (IR Spectrometry test) of one meter box manufactured as per specification is to be conducted at any NABL accredited laboratory/CIPET by the supplier at their own cost after placement of order.

3.16 For type testing the meter box will be selected from the first offered lot of meter with meter box. If the type test results are not found satisfactory, the offered lot of meter along with meter box will be rejected.

4. Quality Control:

- 4.1 Type test reports from CIPET/NABL accredited laboratory as per Standard IS/ASTM shall be submitted. The type test reports shall not be more than 5 (five) years old. Acceptance tests as per IS/ASTM are to be carried out by the supplier in presence of WBSEDCL's representative.

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4.2 Material of meter enclosure shall be tested for Heat Deflection Temperature, Exposure to Flame (Self-extinguishing) and Resistance to Heat & Fire (Glow wire) as per specification and Ref. Standard IS/ASTM.

5. General Construction Requirement:

5.1 Viewing Window: A viewing window as per drawing made up of scratch and break resistant, UV resistant, transparent Polycarbonate material shall be provided on the door for reading the meter without inconvenience. The minimum thickness of the viewing window shall be 2.0 mm (flashing with top). The viewing window shall be provided with fixed rain hood. The window shall be securely fixed with meter enclosure from inside. Suitable neoprene gasket shall be provided so that there shall not be any ingress of moisture through this window into the enclosure.

No viewing window is required for transparent polycarbonate meter box.

5.2 Construction of louver: No louver is required for transparent polycarbonate meter box.

5.3 One push button is to be provided on the front side of the top cover of meter box for taking meter reading during power off condition without opening of meter box cover.

5.4 The box shall be designed in such a way that NIC card shall be kept protected in side the box while shall not hinder data communication.

5.5 Provision for fitting of external antenna shall also be considered during preparation of design of the box.

6. Guarantee: As per NIT

7. Replacement of defective Meter Box: As per NIT

8. Testing:

8.1 Type Test: The bidder must furnish type test report including material verification of the offered/sample meter box from any NABL/Govt. approved laboratory as available with them along with technical bid without which the offer will not be considered. Type test report should not be more than 5 (five) years old.

8.2 Type testing at any recognized NABL accredited laboratory/CIPET in respect of one meter box as per the specified size, selected from any one of the offered lot during supply is to be conducted by the supplier at their own cost after placement of order for verification of material and quality of the box. If the type test results are not found satisfactory, the offered lot of meter along with the meter box will be rejected.

8.3 Acceptance Test: The acceptance test as stipulated in Annexure-VI shall be carried out at the time of inspection of the offered material.

Notes: 1) Where facilities do not exist at supplier's works for carrying out one or more of the Acceptance Tests as per Annexure-VI, such tests may be carried out at any of the approved laboratories such as CIPET/IIT/National Test House/Govt. approved laboratory etc. in presence of WBSEDCL's representative.

8.4 The sampling plan for carrying out the acceptance tests shall be as per IS.

9. Submission of Drawing: Three (3) copies of drawing complete in all respect shall have to be submitted to the Ordering Authority.

10. Inspection: The inspection will be carried out as per inspection & testing clause of General Conditions of Contract (GCC).

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11. Guaranteed Technical Particulars: The bidder shall furnish all the necessary information as per Annexure-VII - Guaranteed Technical Particulars. If the bidder desire to furnish any other information in addition to the details as asked for, the same may be furnished.

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LIST OF TESTS TO BE CARRIED OUT ON SINGLE PHASE METER BOX

Note : Applicable degree of protection shall be IP42 or better. Legend : T- Type Test, R- Routine Test, A- Acceptance Test

Note : Applicable degree of protection shall be IP42 or better. Legend : T- Type Test, R- Routine Test, A- Acceptance Test

ANNEXURE-VI

GUARANTEED TECHNICAL PARTICULARS OF PILFER PROOF METER BOX FOR SINGLE PHASE SMART METER

Sl. No.	Description	Detailed requirement	Offered by the bidder
1.	Material used for moulded meter box	Engineering Plastic (Acrylonitrile Butadiene Styrene)/Polycarbonate	
2.	Grade of Material	Fire Retardant, Self Extinguishing	
3.	Properties of material for meter box		
(a)	Heat Deflection Temperature (Min. 1400 C @ 1.8 MPa) (Ref. Std. IS/ASTM)	1000 C (Minimum for Engg. Plastic) / 1400 C (Minimum for Polycarbonate)	
(b)	Exposure to flame (Ref. Std..IS/ASTM)	Self-extinguishing	
(c)	Melting Point (Ref. Std. IS/ASTM)	1800 C (Minimum for Engg. Plastic) / 2100 C (Minimum for Polycarbonate)	
(d)	Resistance to heat & fire	Glow wire test at 6500 C	
(e)	Mechanical Property		
i)	Tensile Strength (MPa)	To be specified by the bidder	
ii)	Flexural Strength (MPa)	- Do -	
iii)	Modulus of Elasticity (MPa)	- Do -	
4.	Constructional features of the box		
(a)	Clear inside dimensions of Meter Box	Refer Drawing	
i)	Height	To be specified by the bidder	
ii)	Width	- Do -	
iii)	Depth	- Do -	
iv)	Rust & Vermin proofing	Neoprene Rubber Gasket NRG	
(b)	Minimum clearance from meter on all 4 sides	Clearance from all sides of the meter should be 30+2 mm except the bottom side which should be 75+5 mm from the lower edge of terminal block.	
(c)	Minimum clearance from meter on	15+2 mm	

	front		
(d)	Minimum clearance from back of meter	10 + 2 mm	
(e)	Viewing Window :		
	i) Material of transparent cover	Glass/Polycarbonate with Rubber Gasket	
	ii) Size of opening (Min.)	90mm x 75 mm	
	iii) Thickness of moulded sheet	2.0 mm (Minimum)	
	iv) Fixing method	Fixed from inside with rubber gasket	
(f)	Earthing arrangement	May or may not be required.	
(g)	Sealing Arrangement	Holes for wire seal (One No.)	
(h)	Colour of Meter Box (base & cover)	Crystal clear	
(i)	Box mounting arrangement		
	i) No. of holes for fixing of meter box	4 Nos. Holes	
	ii) Dimension of holes	6 mm	
	iii) Dimension of box fixing screw	4X25 mm self threaded	
	iv) Total no. of fixing screws to be provided	4 nos.	
(j)	Hinges	Concealed hinges	
(k)	Incoming & outgoing cable holes	2 Nos. 12 mm. dia. Holes at bottom totally covered with Neoprene Rubber Gasket (NRG)	
(m)	Whether the cover is overlapping type having collars on all four sides	Yes	
(n)	Whether the cover/base provided with semicircular/circular neoprene rubber gasket of 2.5 mm dia (Minm.) to completely fit in the groove of the base	Yes	

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(o)	Whether the meter box having push button to access to the meter for taking reading during power off condition without opening of meter box cover	To be provided	
(p)	Weight of complete box in Kg with +/- tolerance	To be specified by the bidder	
5.	Type test report as per Technical Specification	To be submitted by the bidder	
6.	Degree of protection	IP 42 or better	
7.	Any other information		

A collection of handwritten signatures and initials is present below the table. A prominent diagonal line is drawn across the page, starting from the top center and extending towards the bottom right. The signatures include:

- Top left: "Chin" and "K"
- Top center: "H. J." and "Anwar"
- Top right: "S. B." and "M. B."
- Middle left: "S. B." and "S. B."
- Middle center: "S. B." and "S. B."
- Middle right: "S. B." and "S. B."
- Bottom left: "S. B." and "S. B."
- Bottom center: "S. B." and "S. B."
- Bottom right: "S. B." and "S. B."